

COMPUTER ARCHITECTURE AND ORGANIZATION LABORATORY

Experiment 3

Design and Verification of 4-bit Ripple Carry Adder

and Study of Propagation Delay on Logisim

Experiment No. 3

Subject COA

Branch CSE

Name Anshul Modi

Roll No. 241210021

1. Aim / Objective

To design and verify a 4-bit Ripple Carry Adder (RCA) using basic logic gates on Logisim. The circuit cascades four Full Adder stages so that the carry-out of each stage feeds into the carry-in of the next, and to study the propagation delay introduced by this ripple-carry mechanism across all four stages.

2. Theory

A Ripple Carry Adder adds two n-bit binary numbers by cascading n Full Adders. Each Full Adder receives two operand bits and a carry-in from the previous stage, producing a Sum bit and a Carry-out. The carry propagates ('ripples') from the Least Significant Bit (LSB) to the Most Significant Bit (MSB), introducing cumulative delay.

Key Components:

- **Half Adder:** Adds two 1-bit inputs (A, B). Outputs: Sum ($S = A \oplus B$) and Carry ($C_{out} = A \cdot B$). No carry input.
- **Full Adder:** Adds three 1-bit inputs (A, B, C_{in}). Outputs: Sum ($S = A \oplus B \oplus C_{in}$) and Carry-out ($C_{out} = AB + C_{in}(A \oplus B)$). Implemented with two Half Adders and one OR gate.
- **4-bit RCA:** Four Full Adders cascaded so C_{out} of FA_i drives C_{in} of FA_{i+1} . Inputs: A_3-A_0 , B_3-B_0 , $C_{in} = 0$. Outputs: S_3-S_0 , C_{out} .
- **Propagation Delay:** Total delay $t_{total} = 4 \times t_{FA}$. Worst case: carry ripples through all 4 stages (e.g., $A = 0111$, $B = 0001$).

3. Stage Configuration

The four Full Adder stages are configured as follows within the 4-bit RCA. Each stage processes one bit-position of the 4-bit operands A and B:

Stage	Inputs	Carry-in (C_{in})	Outputs
FA_0 (LSB)	A_0, B_0	0 (fixed)	S_0, C_1
FA_1	A_1, B_1	C_1 from FA_0	S_1, C_2
FA_2	A_2, B_2	C_2 from FA_1	S_2, C_3
FA_3 (MSB)	A_3, B_3	C_3 from FA_2	S_3, C_{out}

4. Full Adder Truth Table

The truth table below defines the operation of each Full Adder stage in the 4-bit RCA:

A	B	C_{in}	Sum (S)	Carry-out (C_{out})
0	0	0	0	0
0	0	1	1	0

0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

5. Procedure

1. Open Logisim and create a new project.
2. Design a **Half Adder** using one XOR gate (Sum) and one AND gate (Carry). Verify outputs against the Half Adder truth table.
3. Design a **Full Adder** by combining two Half Adders and one OR gate. Test all 8 input combinations.
4. Cascade four Full Adder blocks to build the **4-bit Ripple Carry Adder**. Connect C_{out} of each stage to C_{in} of the next.
5. Connect 8 input pins (A3–A0, B3–B0) and set C_{in} of FA_0 to logic 0.
6. Add 5 output pins: S3, S2, S1, S0, and C_{out} .
7. Verify the adder output for multiple input combinations against the expected sum.
8. To study propagation delay: go to **Simulate** → **Tick Frequency** and set a slow rate (2–4 Hz).
9. Enable ticks via **Simulate** → **Enable Ticks** (Ctrl+K). Use **Step Simulation** to observe the carry rippling from FA_0 through FA_3 one stage at a time.
10. Set A = 0111, B = 0001 and step through ticks to observe the worst-case carry ripple across all four stages.
11. Use **Simulate** → **Logging** to add output signals to the chronogram and confirm that later sum bits settle after earlier ones.
12. Record all observations and take screenshots of the completed circuit.

6. Logisim Circuit Screenshot

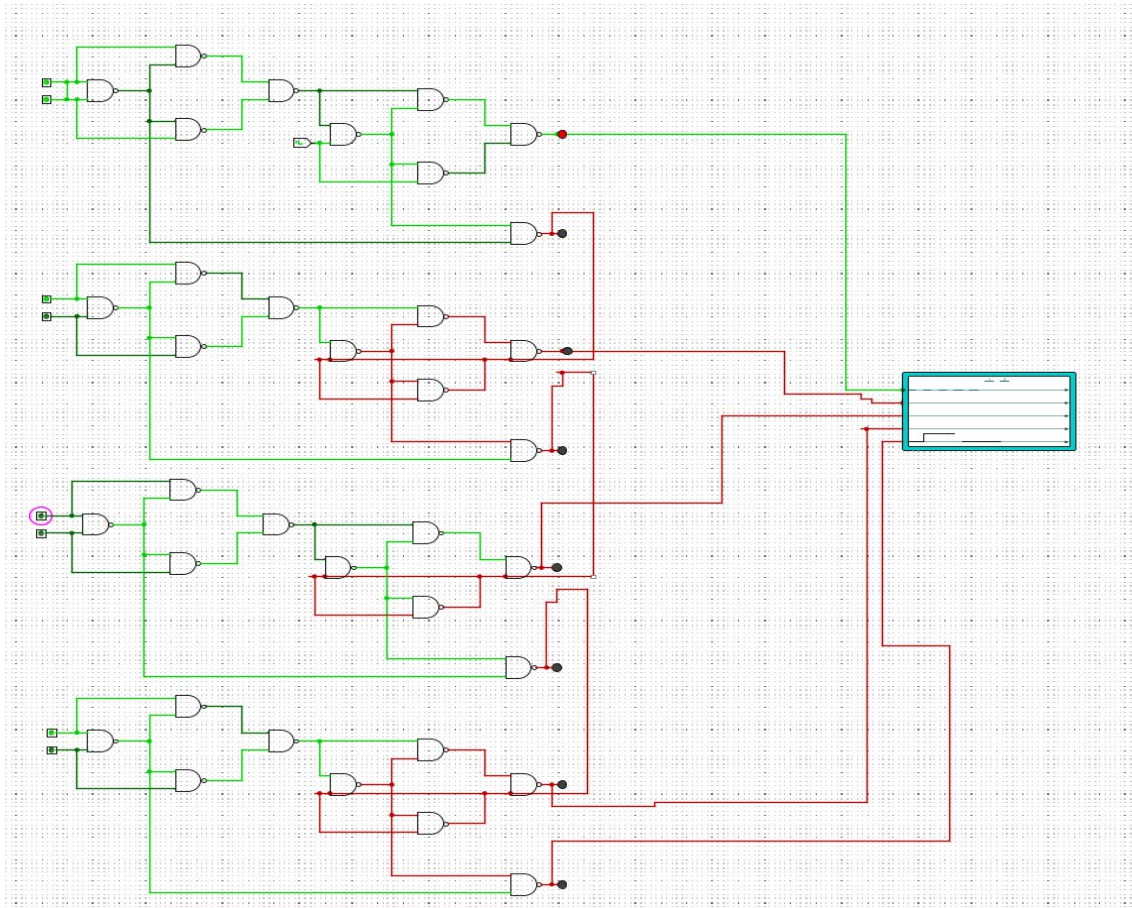


Fig. 1 – 4-bit Ripple Carry Adder on Logisim: Four Full Adder stages (FA_0 – FA_3) cascaded using AND, XOR and OR gates. FA_2 input is circled in pink. Green wires carry data signals; red wires show the carry chain rippling from FA_0 to FA_3 . The chronogram (blue block, right) displays output timing to study propagation delay.

7. Observations

- The 4-bit RCA correctly produced Sum and Carry-out for all tested input combinations, matching the expected theoretical results.
- When $A = 0111$, $B = 0001$ was applied and ticks were stepped slowly, carry was observed to propagate sequentially: $FA_0 \rightarrow FA_1 \rightarrow FA_2 \rightarrow FA_3$.
- Sum bit S_0 settled first, followed by S_1 , S_2 , and S_3 in order, confirming the ripple effect.
- C_{out} was the last signal to stabilize, consistent with maximum propagation delay.
- Worst-case delay was observed for $A = 1111$, $B = 0001$, where carry rippled through all four stages.
- The chronogram (Simulate $\rightarrow$ Logging) visually confirmed that output signals settle at progressively later time steps from LSB to MSB.
- Total propagation delay was confirmed to be proportional to the number of stages: $t_{total} = 4 \times t_{FA}$.

8. Result

A 4-bit Ripple Carry Adder was successfully designed and verified on Logisim using basic logic gates (XOR, AND, OR). All Sum and Carry-out outputs matched the theoretical truth table values. The propagation delay was studied using Logisim's tick-based step simulation and chronogram, confirming that the ripple carry mechanism introduces a cumulative delay of $t_{total} = 4 \times t_{FA}$ in the worst case. The experiment validated both the functional correctness and the inherent timing limitation of the ripple carry

adder architecture.